

高硬钢用2刃微小径球头铣刀
Hardened Steels 2-Flute Micro Diameter Ball Endmills

- 碳素钢 Carbon Steel
- 合金钢 Alloy Steel
- 调质钢 Prehardened Steel
- 高硬度钢 Hardened Steel
- 高硬度钢 Hardened Steel
- 高硬度钢 Hardened Steel

2刃 · 球头2-Flute · Ball

刻模Die-sinking

曲面Profiling

R半径Radius

螺旋Helical

半精加工Semi-Finishing

精加工Finishing

切削条件Cutting Conditions

外径公差Diameter Tolerance

$D \leq 0.1: 0 \sim -0.005$
 $0.15 \leq D \leq 0.9: 0 \sim -0.007$

R公差R Tolerance

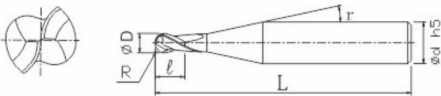
$R \leq 0.2: \pm 0.003$
 $0.25 \leq R \leq 0.45: \pm 0.005$

硬度H5

$0 \sim -0.005$
(mm)



TYPE A :



- ✦ 高硬度材料用球头铣刀。最适用于精加工。
- ✦ 适用于~65HRC的高硬度材料。
- ✦ Ball End Milling Cutter For High Hardness Materials. Most Suitable For Finishing.
- ✦ Suitable For ~65HRC High Hardness Materials.



产品代码 Code No.	规格型号 Spec Typ.	球头半径(R) Radius	刃长(l) Length of Cut	外径(D) Dia.	颈角(r) Neck Taper Angle	柄径(d) Shank Dia.	全长(L) Overall Length	形状 Type	定价 Retail Price
FX6MSB00062	R0.03x0.06L	R0.03	0.06	0.06	11°	4	45	A	
FX6MSB00072	R0.035x0.07L	R0.035	0.07	0.07	11°	4	45	A	
FX6MSB00082	R0.04x0.08L	R0.04	0.08	0.08	11°	4	45	A	
FX6MSB00092	R0.045x0.09L	R0.045	0.09	0.09	11°	4	45	A	
FX6MSB0012	R0.05x0.1L	R0.05	0.1	0.1	11°	4	45	A	
FX6MSB00122	R0.06x0.12L	R0.06	0.12	0.12	11°	4	45	A	
FX6MSB00142	R0.07x0.14L	R0.07	0.14	0.14	11°	4	45	A	
FX6MSB00152	R0.075x0.15L	R0.075	0.15	0.15	11°	4	45	A	
FX6MSB00162	R0.08x0.16L	R0.08	0.16	0.16	11°	4	45	A	
FX6MSB00182	R0.09x0.18L	R0.09	0.18	0.18	11°	4	45	A	
FX6MSB0022	R0.1x0.2L	R0.1	0.2	0.2	15°	4	45	A	
FX6MSB0032	R0.15x0.3L	R0.15	0.3	0.3	15°	4	45	A	
FX6MSB0042	R0.2x0.6L	R0.2	0.6	0.4	15°	4	45	A	
FX6MSB0052	R0.25x0.8L	R0.25	0.8	0.5	15°	4	45	A	
FX6MSB0062	R0.3x0.9L	R0.3	0.9	0.6	15°	4	45	A	
FX6MSB0082	R0.4x1.2L	R0.4	1.2	0.8	15°	4	45	A	
FX6MSB0092	R0.45x1.4L	R0.45	1.4	0.9	15°	4	45	A	

高硬度材料加工用。超群的切削排出性能，可实现高效率加工。
For machining of high-hardness materials. Excellent chip removal performance enables high efficiency.

切削参数参考表
Recommended Milling Conditions

- 碳素钢 Carbon Steel
- 合金钢 Alloy Steel
- 调质钢 Prehardened Steel
- 高硬度钢 Hardened Steel
- 高硬度钢 Hardened Steel
- 高硬度钢 Hardened Steel

加工材料 Work Material	高速钢 Hardened Steels SKH (~65HRC)				淬火钢 Hardened Steels SKD11 (~62HRC)				淬火钢 Hardened Steels SKD61 • STAVAX • HPM-38 (~52HRC)			
	主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut	
	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
球头半径(R) Radius												
0.03	40,000	80	0.001	0.001	40,000	120	0.001	0.001	40,000	100	0.004	0.008
0.035	40,000	100	0.001	0.002	40,000	140	0.002	0.002	40,000	150	0.004	0.008
0.04	40,000	120	0.001	0.002	40,000	160	0.004	0.004	40,000	200	0.004	0.008
0.045	40,000	140	0.001	0.003	40,000	180	0.004	0.004	40,000	220	0.006	0.01
0.05	40,000	140	0.001	0.003	40,000	200	0.006	0.006	40,000	240	0.006	0.01
0.06	40,000	160	0.002	0.004	40,000	220	0.006	0.006	40,000	260	0.008	0.015
0.07	40,000	160	0.002	0.004	40,000	240	0.008	0.008	40,000	280	0.008	0.015
0.075	40,000	180	0.002	0.004	40,000	260	0.008	0.008	40,000	300	0.01	0.02
0.08	40,000	200	0.003	0.005	40,000	280	0.01	0.01	40,000	320	0.01	0.02
0.09	40,000	220	0.003	0.005	40,000	300	0.01	0.01	40,000	340	0.01	0.02
0.1	40,000	240	0.003	0.005	40,000	320	0.01	0.01	40,000	360	0.01	0.02
0.15	40,000	300	0.005	0.01	40,000	360	0.01	0.02	40,000	420	0.01	0.03
0.2	40,000	480	0.01	0.02	40,000	820	0.02	0.05	40,000	1,000	0.02	0.06
0.25	40,000	600	0.015	0.03	40,000	1,000	0.025	0.05	40,000	1,200	0.03	0.07
0.3	30,000	720	0.02	0.05	40,000	1,200	0.03	0.06	40,000	1,600	0.05	0.1
0.4	30,000	1,200	0.05	0.1	40,000	1,800	0.07	0.1	40,000	2,200	0.1	0.15
0.45	27,500	1,300	0.06	0.1	35,000	1,900	0.08	0.15	40,000	2,350	0.1	0.2
备注 Notes	※切深量的ap表示轴向切入量,ae表示步距量。 ※建议使用油雾冷却方式。 ※请以相同的比率调整主轴转速和进给速度。 ※加工参数会因切深量和机床刚性的状况而有所不同。请每次调整后在使用。 ※请根据需要控制刀具的伸出量。 ※ Depth of Cut: ap=Axial Depth of Cut / ae=Radial Depth of Cut. ※ We recommend using oil mist coolant. ※ Adjust both spindle speed and feed at the same rate. ※ Adjust milling conditions according to the volume of depth of cut and rigidity of machine. ※ Length of tool overhang must be as short as possible.											